INCH-POUND
MIL-DTL-24640/11E
8 November 2011
SUPERSEDING
MIL-DTL-24640/11D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-TRIAD, 18 AWG, TYPES 3XS AND 3XSW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (3XS) and Watertight (3XSW)

First	-	19-strand, tin-coated copper conductor, 18 AWG. Watertight constructions shall be constructed with 7 strands.
Second	-	Component wires shall be SAE-AS81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed in this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Three conductors, one black, one white, and one red, twisted together. The pitch shall be 2±¼ inch.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield of 38 AWG tin-coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.
Sixth	-	Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2.
Seventh	-	The required number of shielded, twisted triads shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Eighth	-	Optional binder tape.
Ninth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

PIN M24640/11	Type number	Number of triads	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	3XS-7	3	0.044	0.601	0.647	315.0
-02UN	3XSW-7	3	0.044	0.600	0.645	350.0

REQUIREMENTS:

Qualification required.

Construction details: See table I.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	
Non-blocked	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Water-blocked	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	
Non-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 168 hours
Water-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours
Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)

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Shrinkage	
Non-blocked	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Water-blocked	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	6.67
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor and shield	1000
Component shield to shield	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield)	Not applicable
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	55
Capacitance unbalance (percent, max.)	15
Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (type 3XSW only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Drip (type 3XSW only)	No failure
Cross-link proof test (percent, max.)	

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Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, min.)	Not applicable
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (type 3XSW only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable
Group D:	
Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-079)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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